Examiner only

2. Iron is extracted from iron oxide inside the blast furnace.

(a)

iron ore	**steel**	**limestone**	**hot air**
slag	**iron**	**coke**	**waste gases**

Choose substances from the box to complete the following sentences. [2]

Iron oxide, coke and .... are the materials fed in at the **top** of the furnace.

The furnace is heated by burning the coke in ....

(b) Balance the symbol equation that represents the main reaction taking place inside the furnace. [1]

$$Fe_2O_3 + 3CO \longrightarrow \boxed{}\,Fe + 3CO_2$$

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(c) Some of the impurities are removed from the furnace as slag. Slag contains calcium silicate which has the chemical formula $CaSiO_3$.

Tick (✓) the box next to the calculation used to find the correct relative formula mass of calcium silicate, $CaSiO_3$. [1]

$A_r(\text{Ca}) = 40$ $A_r(\text{Si}) = 28$ $A_r(\text{O}) = 16$

40 + 28 + 16 ☐

40 + 40 + 40 + 28 + 28 + 28 + 16 + 16 + 16 ☐

40 + 28 + 16 + 16 + 16 ☐

40 + 16 + 16 + 16 ☐

40 + 28 + 28 + 28 + 16 + 16 + 16 ☐

(d) 820 tonnes of iron was extracted from 1 750 tonnes of iron ore. Calculate the percentage of iron in the ore. Give your answer to **one** decimal place. [2]

Percentage = ... %

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Turn over.